FIKR Model — Lesson Plan

Learning Leading to Production

Title	Dividing Polynomials	**Subject / Unit**	Topic 3 — Polynomial Functions · Lesson 3-4
Grade	Grade 10 — Algebra II	**Date**	Week 10 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

Not shown — the Scope and Sequence document has no Essential Question column, so it is left open rather than invented. Add it when confirmed.

Lesson Objectives (curriculum map — verbatim)

1. Identify quotients and remainders upon the division of polynomials.
2. Apply the Remainder Theorem and the Factor Theorem.
3. Use synthetic division to divide a polynomial by a linear polynomial with leading coefficient 1.

Standards (curriculum map — verbatim)

HSA.APR.B.2 Know and apply the Remainder Theorem: For a polynomial p(x) and a number a, the remainder on division by x - a is p(a), so p(a) = 0 if and only if (x - a) is a factor of p(x).

HSA.APR.D.6 Rewrite simple rational expressions in different forms; write a(x)/b(x) in the form q(x) + r(x)/b(x), where a(x), b(x), q(x), and r(x) are polynomials with the degree of r(x) less than the degree of b(x), using inspection, long division, or, for the more complicated examples, a computer algebra system.

HSA.SSE.A.2 Use the structure of an expression to identify ways to rewrite it.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: Divisor ; factor ; Polynomial ; long division ; proper rational function ; Quotient ; Remainder ; synthetic division.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

0 Preparation Before Class

Learning Outcome: What will the student be able to do?

Students will identify the quotient and the remainder when one polynomial is divided by another (objective 1; HSA.APR.D.6).

Students will apply the Remainder Theorem to find $p(a)$ as a remainder and the Factor Theorem to test a factor (objective 2; HSA.APR.B.2).

Students will use synthetic division to divide a polynomial by $x - a$, including a placeholder zero for a missing

power (objective 3).

Students will rewrite $\frac{a(x)}{b(x)}$ as $q(x) + \frac{r(x)}{b(x)}$ (HSA.APR.D.6; HSA.SSE.A.2).

Three teaching days this week. If time runs short, cut Smart Production — never the Mastery Gate.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Pre-session home journey (already issued, with Lesson 3-3): watch "Remainder Theorem and Synthetic Division of Polynomials", study two worked examples, answer three ungraded readiness questions. Submitted on the LMS before class.

Readiness questions: divide $x^3 - 3x^2 + x - 5$ by $x - 2$; find $p(1)$ for a cubic; decide whether $x - 2$ is a factor.

Evidence: the LMS answers feed the Diagnose.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — the slide-4 checks: evaluate $p(x) = x^3 - 4x + 2$ at $x = 3$; factor $x^2 - x - 12$; list the coefficients of $2x^3 - 5x + 1$ with the missing power.

Answers: 17; $(x - 4)(x + 3)$; $2,\ 0,\ -5,\ 1$.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gaps — Q1: 3^3 evaluated as 9. Q3: the zero for x^2 is left out.

Routing: 0–1 correct → re-teach substituting and the placeholder zero. 2 → straight to synthetic division. 3 → begin Investigate early.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 5 — Long division (objective 1; HSA.APR.D.6). $x^2 + 5x + 7 = (x + 2)(x + 3) + 1$, so the quotient is $x + 3$ and the remainder is 1; $\frac{2x^3 - 3x^2 + 4x - 5}{x - 1} = 2x^2 - x + 3 + \frac{-2}{x - 1}$.

Slide 6 — Synthetic division (objective 3). $x^3 + 2x^2 - 4x + 5$ by $x - 1$ gives $x^2 + 3x - 1$ remainder 4; $x^3 - 4x^2 + x + 6$ by $x - 2$ gives $x^2 - 2x - 3$; $2x^3 + x^2 - 13x + 6$ by $x + 3$ uses the box number -3.

Slide 7 — Remainder and Factor Theorems (objective 2; HSA.APR.B.2). $p(3) = 5$ for $x^3 - 3x^2 + x + 2$; $x + 1$ is a factor of $x^3 + 2x^2 - x - 2$, which factors as $(x + 1)(x + 2)(x - 1)$; $k = -1$ makes $x - 2$ a factor of $x^3 + kx^2 - 4x + 4$.

Slide 8 — Rewriting a rational expression (HSA.APR.D.6). $\frac{x^2 + 3x + 5}{x + 1} = x + 2 + \frac{3}{x + 1}$ by inspection;

$\frac{x^3+8}{x+2} = x^2 - 2x + 4$, linking to the sum-of-cubes identity from Lesson 3-3.

Narration focus — for $x - a$ write a; a missing power needs a zero; the last number is the remainder.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: divide $x^3 + 2x^2 - 5x + 1$ by $x - 1$ with synthetic division.

Expected: quotient $x^2 + 3x - 2$, remainder -1. Watch for the box number -1 and for forgetting the remainder.

80% correct → release guided practice.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two identical "Give it a go" problems, worked with the teacher:

(1) Divide $x^3 - 2x^2 - 5x + 6$ by $x - 3$. (2) Find the remainder when $x^3 - 5x + 7$ is divided by $x - 2$.

Answers: (1) quotient $x^2 + x - 2$, remainder 0. (2) $p(2) = 5$.

Worked live; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

"Choose Your Route" sheet with live Pear Deck feedback. Routes name the task, not the student.

PRACTICE — Secure the method. Six cards: synthetic division, the Remainder Theorem and a factor test; then find the error in a box number. Done when: the box number has the opposite sign to the constant in the divisor and a zero remainder is linked to a factor.

APPLY — Use it in context. A box volume, a value of k, a rational expression and a value of a. Done when: the divisor is written as $x - a$ first and each answer is checked.

INVESTIGATE — Find out why. Prove $r = p(a)$, factor a cubic fully, and find remainders without dividing. Done when: the argument starts from $p(x) = (x - a)q(x) + r$.

Page 3: compare three division routes, judge three claims, test a What if...? idea. Every student sits the same Mastery Gate.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 11).

A model Qiddiya water-slide tank has volume $V(x) = x^3 + 7x^2 + 14x + 8$ m^3 and height $(x + 2)$ m. The model values are invented for practice.

(a) Show that $x + 2$ is a factor and find the base area. (b) Factor the base area and write $V(x)$ as three factors.

(c) For $x = 3$ find the volume from the factors and check it from $V(x)$.

Answers: (a) $V(-2) = 0$; base area $x^2 + 5x + 4$. (b) $(x + 1)(x + 4)$, so $V = (x + 1)(x + 2)(x + 4)$. (c) $4 \times 5 \times 7 = 140$ m^3; check $27 + 63 + 42 + 8 = 140$.

Students then hand their reasoning to the AI challenger.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum standard: the box number, the coefficients, the remainder read, the factors written, units (m, m^3).

Role of AI: challenger only — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently: the written justification for (a)–(c).

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Divide $x^3 + 4x^2 - x - 6$ by $x + 1$. 2. Is $x + 2$ a factor of $x^3 + x^2 - 4x - 4$? 3. Find k so that $x - 2$ is a factor of $x^3 + kx - 6$.

Answers: 1. quotient $x^2 + 3x - 4$, remainder -2. 2. $p(-2) = 0$, so yes. 3. $8 + 2k - 6 = 0$, so $k = -1$.

Done when: question 1 reads the remainder with its sign and question 2 states the Factor Theorem.

Scored live in Pear Deck and logged in the Impact Dashboard.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise their tank reasoning using the AI challenger's feedback and the teacher's micro-conference note.

Final product (Canva one-pager, photographed solution, or 45-second voice note): the division, the Factor Theorem used, ONE non-example (box number 3 for $x + 3$), and why the three factors give the dimensions.

Graded on clarity of reasoning.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall.

Total minutes across the seven stages: **60** (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars

Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Six cards, a find-the-error line and a synthetic frame. Done when: the box number has the opposite sign.

APPLY — Use it in context. A box volume, a value of k, a rational expression and a value of a. Done when: each answer is checked.

INVESTIGATE — Find out why. Prove $r = p(a)$, factor a cubic and find remainders without dividing. Done when: the argument starts from $p(x) = (x - a)q(x) + r$.

No student is labelled; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

Qiddiya water-slide tank and a model box; invented values.

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Advanced Math: when $p(x) = x^3 - 2x^2 + kx - 1$ is divided by $x - 3$ the remainder is 5, so $k = -1$. Trap: using $p(-3)$.

GAT (Qudurat) — the remainder of $x^2 + 1$ divided by $x - 3$ against 10: they are equal. Trap: assuming the divisor decides the size.

SAAT (Tahsili) — the quotient of $x^3 - 8$ divided by $x - 2$ is $x^2 + 2x + 4$. Trap: $x^2 - 2x + 4$.

Suggestions for Improvement: